



## MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-PH101/BSPH101 Physics-I

UPID : 001003

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

## Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[ 1 x 10 = 10 ]

- (i) What is the emissivity of an ideal black body?
- (ii) What is the dimension of phase space?
- (iii) The Curl of a conservative vector field is \_\_\_\_\_.
- (iv) In Fraunhofer diffraction, the incident wavefront is \_\_\_\_\_.
- (v) The displacement current arises due to \_\_\_\_\_.
- (vi) For an ideal blackbody, the power emitted per unit area is proportional to \_\_\_\_\_.
- (vii) At absolute zero temperature what is the probability of occupancy of an electron in an energy state above the Fermi level?
- (viii) Find the angle between the vectors  $\hat{i}$  and  $2\hat{i} + \hat{j}$
- (ix) Diffraction is actually a form of interference. True or False?
- (x) Dimension of polarizability in SI units is \_\_\_\_\_.
- (xi) If  $E_1$  be the energy of the ground state of a 1 dimensional potential well of length  $L$  and  $E_2$  be the energy of the ground state of a 1 dimensional potential well of length  $L/2$ , what is the relation between  $E_1$  and  $E_2$ ?
- (xii) What is the average energy of electrons in a metal at 0 Kelvin?

## Group-B (Short Answer Type Question)

Answer any three of the following :

[ 5 x 3 = 15 ]

2. State five applications of LASER [5]
3. Mention the criteria for the applicability of Bose-Einstein statistics. [5]
4. Find the value of the given Commutator and explain the significance of the result [5]

$$[\hat{x}, \hat{p}_x]$$

5. What do you mean by Macrostate and Microstate? –Explain with suitable example. [5]
6. Compute the smallest possible uncertainty in the position of an electron moving with speed 30000000 metres per second. [5]

## Group-C (Long Answer Type Question)

Answer any three of the following :

[ 15 x 3 = 45 ]

7. (a) Give one example of Solid, Liquid and Gaseous Dielectric material [ 3 ]
- (b) State three applications of Dielectric Materials [ 3 ]
- (c) What is Dielectric Constant? What is its value in vacuum and conductor? [ 2+1+1 ]
- (d) A capacitor is made up of a dielectric material of  $\epsilon_r=4.2$ . It has an effective surface area of  $0.09\text{m}^2$  and capacitance of  $10\mu\text{F}$ . Calculate the field strength if a potential difference of 5V exists across the capacitor. Given  $\epsilon_0=8.85 \times 10^{-12} \text{ Fm}^{-1}$  [ 5 ]
8. (a) State the Planck's radiation law explaining all terms and show that the law reduces to the Wien's radiation law and Rayleigh Jeans law in opposite limits. [ 2+2+2 ]
- (b) Define group velocity and Phase velocity. Show that the product of group velocity and phase velocity is a constant. [ 2+3 ]

- (c) Write down the time dependent Schrodinger equation clearly mentioning the terms. Explain the physical significance of a wave function [ 2+2 ]
9. (a) A particle is executing simple harmonic oscillation and has a velocity of 5m/s when the displacement is 2m and reduces to 2m/s when the displacement is 5m. Calculate [ 4+2+2 ]
- Amplitude
  - Frequency
  - Time period
- of the above oscillation
- (b) Define holonomic and non holonomic Constraints with examples of each [ 2+2 ]
- (c) What is a conservative vector field? Give one example [ 2+1 ]
10. (a) Compare diamagnetic, paramagnetic and ferromagnetic materials [ 6 ]
- (b) Differentiate between Soft and Hard magnetic materials [ 3 ]
- (c) The value of permeability of a material is 0.12 N/A<sup>2</sup>. What is the relative permeability and magnetic susceptibility? [ 2+2 ]
- (d) Define magnetic domain of ferromagnetic materials. [ 2 ]
11. (a) Find the minimum number of lines of a grating which can resolve in the second order spectrum of two lines having wavelengths 5890 Angstrom and 5896 Angstrom. [ 5 ]
- (b) Two polarisers are placed at crossed position. At what angle should a third polariser be placed so that the intensity of the emergent light is one-fourth of the intensity of incident light? [ 5 ]
- (c) What is the highest order spectrum which may be observed with monochromatic light of wavelength 200 nm by using a grating with 2000 lines/cm [ 5 ]

\*\*\* END OF PAPER \*\*\*